Pearson Edexcel International A Level · Pure Mathematics 2

Chapter 6 — Trigonometric identities and equations
Original Practice Questions

30 complete questions · Questions followed by answers and suggested mark schemes

- Answer all parts and show sufficient working.
- Give exact answers unless a degree of accuracy is requested.
- Angles are in radians unless degrees are explicitly indicated.
- Diagrams are not necessarily drawn to scale.
- The marks for each subpart are shown in brackets.

P2 第6章 · 三角恒等式与方程
英文题目 · 中文解析 · 建议评分细则
本册30题；P1、P2、P3全套25章共750题。

内部复核稿，未完成外部教师逐题审阅。
Original practice material. Not a Pearson examination paper.

1.

For all angles for which both sides are defined,

(a) prove that $\dfrac{1-\cos^2 x}{\sin x} = \sin x$. (3)

(b) State all excluded values in $0^\circ \leq x \leq 360^\circ$. (2)

(Total for Question 1 is 5 marks)

For $0^\circ \leq x < 360^\circ$, $2\sin^2 x + \sin x - 1 = 0$.

(a) Find all exact solutions. (4)

(b) State how many solutions lie in $180^\circ < x < 360^\circ$. (1)

(Total for Question 2 is 5 marks)

$3\cos^2 x - 5\cos x + 2 = 0$, where $0 \leq x \leq 2\pi$.

(a) Find all solutions, giving non-exact angles to 3 decimal places in radians. (4)

(b) Find the least strictly positive solution. (2)

(Total for Question 3 is 6 marks)

4.

For $0^\circ \leq x < 360^\circ$, $\sin x \cos x = \frac{1}{2} \sin x$.

(a) Solve the equation without losing any solutions. (4)

(b) Identify the solutions lost by dividing the original equation by $\sin x$. (2)

(Total for Question 4 is 6 marks)

5.

For $-\pi < x \leq \pi$, $\tan x = \sqrt{3}$.

(a) Find every exact solution. (3)

(b) For each solution, state the signs of $\sin x$ and $\cos x$. (2)

(Total for Question 5 is 5 marks)

For $0^\circ \leq \theta \leq 180^\circ$, $\sin(2\theta) = \frac{\sqrt{3}}{2}$.

(a) Find all values of θ. (3)

(b) Find all solutions if the domain is extended to $0^\circ \leq \theta \leq 360^\circ$. (3)

(Total for Question 6 is 6 marks)

7.

Prove the following identities wherever both sides are defined.

(a) $\sin x \tan x + \cos x = \dfrac{1}{\cos x}$. (3)

(b) $\dfrac{\sin x}{1 - \cos x} = \dfrac{1 + \cos x}{\sin x}$. (3)

(Total for Question 7 is 6 marks)

8. P2-C06-R08 · P2 · Chapter 6

For $0 \leq x < 2\pi$, $2\cos^2 x = 3\sin x$.

(a) Rewrite the equation as a quadratic in $\sin x$. (2)

(b) Hence find all exact solutions. (4)

(Total for Question 8 is 6 marks)

9. P2-C06-R09 · P2 · Chapter 6

For $0^\circ \leq x \leq 360^\circ$, $\cos(3x) = 0$.

(a) Find all solutions. (3)

(b) State the spacing between successive solutions and explain it using the input factor 3. (2)

(Total for Question 9 is 5 marks)

$\sin x = \cos x$ for $-180^\circ \leq x \leq 180^\circ$.

(a) Find all solutions. (3)

(b) Find the exact common value of $\sin x$ and $\cos x$ at each solution. (3)

(Total for Question 10 is 6 marks)

For $0 \leq x \leq \pi$, $4\sin^2 x = 3$.

(a) Find the exact solutions. (3)

(b) Find all solutions in $-\pi \leq x \leq \pi$. (2)

(Total for Question 11 is 5 marks)

The expression $F(x) = \dfrac{1 - \sin x}{\cos x}$ is defined where $\cos x \neq 0$.

(a) Show that $F(x) = \dfrac{\cos x}{1 + \sin x}$ on their common domain. (3)

(b) Find the exact value of $F(30^\circ)$. (3)

(Total for Question 12 is 6 marks)

For $0^\circ \leq \theta < 360^\circ$, $\tan^2 \theta = 3$.

(a) Find all exact solutions. (3)

(b) Find the solutions that also satisfy $\cos \theta < 0$. (2)

(Total for Question 13 is 5 marks)

The graph of $y = \sin x$ and the horizontal line $y = k$ are considered for $0 \leq x \leq 2\pi$.

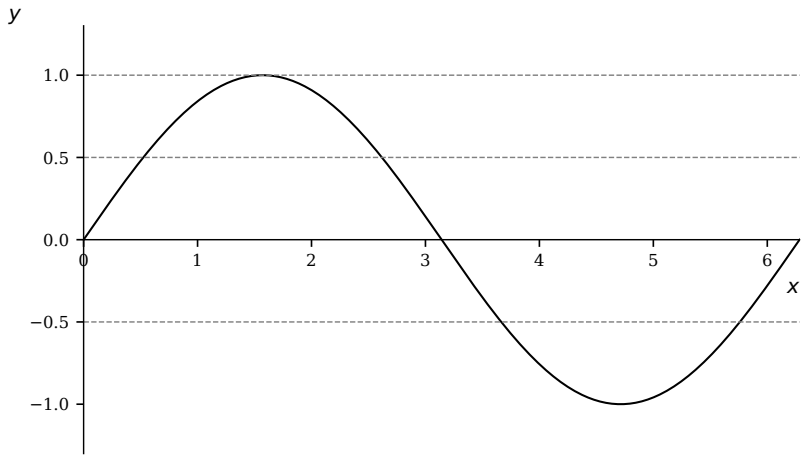


Figure 14

(a) State the number of distinct intersections when $k = 0$, $k = 1/2$ and $k = 1$. (3)

(b) Find every real k giving exactly two distinct intersections. (3)

(Total for Question 14 is 6 marks)

For $0^\circ \leq x \leq 180^\circ$, $\cos(2x - 30^\circ) = 1/2$.

(a) Find all solutions. (4)

(b) Find the least positive solution if the interval is extended to all $x > 0$. (2)

(Total for Question 15 is 6 marks)

For all x for which it is defined, $H(x) = \dfrac{\tan^2 x}{1 + \tan^2 x}$.

(a) Show that $H(x) = \sin^2 x$. (3)

(b) Solve $H(x) = 1/4$ for $0^\circ \leq x < 360^\circ$. (4)

(Total for Question 16 is 7 marks)

For $0 \leq x < 2\pi$, $\cos^2 x = \sin^2 x + 1/2$.

(a) Find $\sin^2 x$. (2)

(b) Hence find all exact solutions. (3)

(Total for Question 17 is 5 marks)

A point on a circle has coordinates $P = (5\cos\theta, 5\sin\theta)$, with $0^\circ \leq \theta < 360^\circ$.

(a) Show that $x^2 + y^2 = 25$ for every such point. (2)

(b) The point also lies on $y = \frac{5}{2}$. Find

(i) all possible values of θ, (2)

(ii) the exact corresponding x-coordinates. (2)

(Total for Question 18 is 6 marks)

The equation $\sin^2 x = \sin x$ is considered.

(a) Find all solutions in $0^\circ \leq x \leq 360^\circ$. (4)

(b) A student replaces the equation by $\sin x = 1$. State the missing solutions and the incorrect operation used. (2)

(Total for Question 19 is 6 marks)

For $-\pi/2 < x < \pi/2$, $3\tan x + 1 = 0$.

(a) Find x in radians to 3 decimal places. (3)

(b) Find the exact value of $\cos x$. (3)

(Total for Question 20 is 6 marks)

21.

For $0^\circ \leq x < 360^\circ$, $4\sin x = 5\cos^2 x$.

(a) Find the exact possible values of $\sin x$. (4)

(b) Find all solutions x to 1 decimal place. (2)

(Total for Question 21 is 6 marks)

The graphs $y = \cos x$ and $y = 2\sin^2 x - 1$ are considered on $0 \leq x \leq 2\pi$.

(a) Find their intersection x-coordinates exactly. (4)

(b) Find the y-coordinate at each intersection. (2)

(Total for Question 22 is 6 marks)

For $0^\circ \leq x \leq 360^\circ$, $\sin(3x + 20^\circ) = 0$.

(a) Find all solutions, giving exact fractional degree values where needed. (4)

(b) Find the number of these solutions in $0^\circ \leq x \leq 180^\circ$. (2)

(Total for Question 23 is 6 marks)

Given $\sin\alpha = 3/5$ and $90^\circ < \alpha < 180^\circ$,

(a) find exactly

(i) $\cos\alpha$ and $\tan\alpha$, (3)

(ii) $\dfrac{1+\sin\alpha}{1-\sin\alpha}$. (2)

(b) Find the exact value of $\sin(180^\circ - \alpha)$ using the graph symmetry of sine. (2)

(Total for Question 24 is 7 marks)

The expression $G(x) = \sin^2 x + 4\cos^2 x$ is defined for all real x.

(a) Show that $G(x) = 1 + 3\cos^2 x$ and find its minimum and maximum possible values. (3)

(b) Find all x in $0^\circ \leq x < 360^\circ$ for which $G(x) = 7/4$. (3)

(Total for Question 25 is 6 marks)

For $0 \leq x \leq 2\pi$, $\tan x = 2\sin x$, with $\cos x \neq 0$.

(a) Find all exact solutions. (4)

(b) State the number of solutions if both endpoints are excluded. (2)

(Total for Question 26 is 6 marks)

27.

The equation $2\cos^2 x + \cos x = 0$ holds for $0^\circ \leq x < 360^\circ$.

(a) Find all solutions. (4)

(b) Find the sum of the solutions in degrees. (2)

(Total for Question 27 is 6 marks)

The graphs $y = \sin(2x)$ and $y = 1/2$ are considered for $0 \leq x \leq \pi$.

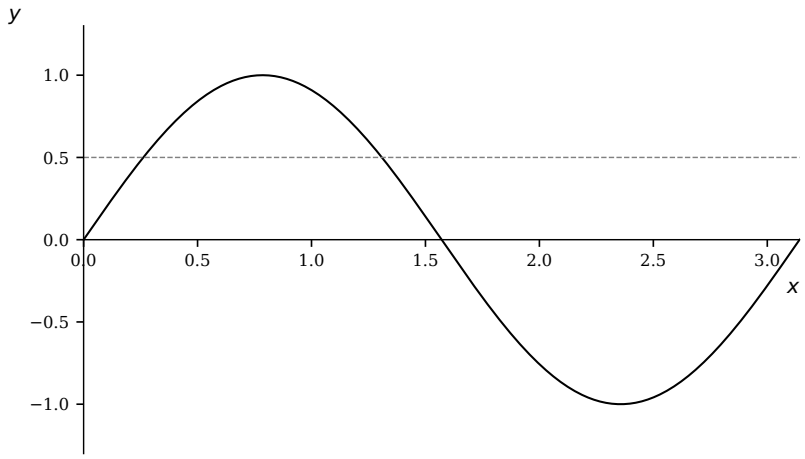


Figure 28

(a) Find the exact intersection coordinates. (3)

(b) Find the total length of the set of x-values for which $\sin(2x) \geq 1/2$ in this interval. (3)

(Total for Question 28 is 6 marks)

An angle x satisfies $0^\circ < x < 90^\circ$ and $\sin x + \cos x = 7/5$.

(a) Find $\sin x \cos x$ exactly. (3)

(b) Find the two possible values of $\sin x$ and hence the two possible values of x to 1 decimal place. (4)

(Total for Question 29 is 7 marks)

For $0 \leq \theta \leq 2\pi$, $5\sin^2\theta + \cos\theta = 3$.

(a) Find the possible exact values of $\cos\theta$. (3)

(b) Find all solutions to 3 decimal places in radians. (4)

(Total for Question 30 is 7 marks)

Answers and Suggested Mark Schemes

中文解答与建议评分点

M：方法分；A：准确分，通常依赖对应方法；B：独立得分点。
等价正确解法应获相应分数。评分点为原创建议，并非官方评分细则。

Question 1

P2-C06-R01 · 5 marks

(a) · 3 marks

$1-\cos^2 x=\sin^2 x$, so the quotient equals $\sin x$ when $\sin x \neq 0$.

用基本平方恒等式，除法只在正弦不为0时合法。

M1	平方恒等式
M1	合法约分
A1	目标式

(b) · 2 marks

$x=0^\circ, 180^\circ, 360^\circ$.

原分母$\sin x=0$处均排除；不能因化简后有值就补回。

M1	原分母零点
A1	全部角

Question 2

P2-C06-R02 · 5 marks

(a) · 4 marks

$x = 30^\circ, 150^\circ, 270^\circ$.

(2sinx−1)(sinx+1)=0，所以sinx=1/2或−1，分别列角。

M1	正确二次分解
M1	两种三角值
M1	对应全部角
A1	完整集合

(b) · 1 marks

One, 270°.

筛选(a)中的值，端点未产生额外解。

B1	数量和角

Question 3

P2-C06-R03 · 6 marks

(a) · 4 marks

$x = 0, \arccos(2/3), 2\pi - \arccos(2/3), 2\pi$, approximately $0, 0.841, 5.442, 2\pi$.

分解(3cosx−2)(cosx−1)；cos=1在两个包含的端点都取到。

M1	正确分解
M1	余弦两支
M1	端点完整
A1	弧度精度

(b) · 2 marks

$\arccos(2/3) \approx 0.841$.

0不严格正，按从小到大选择第二项。

M1	正确排除0
A1	最小正角

Question 4

P2-C06-R04 · 6 marks

(a) · 4 marks

$x = 0^\circ, 60^\circ, 180^\circ, 300^\circ$.

因式sinx(cosx−1/2)=0；不应直接除sinx而丢0、180。

M1	因式分解
M1	两组条件
M1	全部象限
A1	解集

(b) · 2 marks

0° and 180°.

除以某因子将剔除该因子零点，需回原式验证这两点确为解。

M1	原式核查零因子
A1	两角

Question 5

P2-C06-R05 · 5 marks

(a) · 3 marks

$x = -2\pi/3, \pi/3$.

正切周期π，参考角π/3；从π/3+kπ筛选指定区间。

M1	周期解
M1	区间筛选
A1	两精确角

(b) · 2 marks

Both negative at $-2\pi/3$; both positive at $\pi/3$.

负角−120°同终边240°为第三象限；π/3第一象限。

M1	象限定位
A1	两组符号

Question 6

P2-C06-R06 · 6 marks

(a) · 3 marks

$30^\circ, 60^\circ$.

输入2θ走0°到360°，正弦为√3/2在60°和120°，再除2。

M1	扩大输入范围
M1	全部输入解
A1	θ值

(b) · 3 marks

$30^\circ, 60^\circ, 210^\circ, 240^\circ$.

输入范围0°到720°，加一周期的420°、480°后除2。

M1	处理扩展输入范围
M1	周期补解
A1	完整集合

Question 7

P2-C06-R07 · 6 marks

(a) · 3 marks

$\sin x \tan x + \cos x = (\sin^2 x + \cos^2 x)/\cos x = 1/\cos x$.

把tan写成sin/cos再通分，用平方和1。

M1	代入正切商
M1	通分平方和
A1	目标式

(b) · 3 marks

Multiply the left numerator and denominator by $1 + \cos x$ and use $1 - \cos^2 x = \sin^2 x$.

共轭乘式将分母化为$\sin^2 x$，再约去一个$\sin$；保留原域限制。

M1	共轭乘式
M1	平方恒等式与约分
A1	恒等式

Question 8

P2-C06-R08 · 6 marks

(a) · 2 marks

$2\sin^2 x + 3\sin x - 2 = 0.$

$\cos^2=1-\sin^2$，代入后移项。

M1	正确平方恒等式
A1	二次式

(b) · 4 marks

$x = \pi/6, 5\pi/6.$

$(2\sin x-1)(\sin x+2)=0$；$\sin=-2$非法，只留$1/2$。

M1	因式分解
M1	排除范围外三角值
M1	对应角
A1	解集

Question 9

P2-C06-R09 · 5 marks

(a) · 3 marks

$30^\circ, 90^\circ, 150^\circ, 210^\circ, 270^\circ, 330^\circ$.

3x在0°到1080°，余弦零点90°+180°k共六个，除3。

M1	正确输入区间
M1	列齐输入零点
A1	六x

(b) · 2 marks

60°; cosine zeros are 180° apart in input, so x spacing is $180^\circ/3$.

区别函数周期120°与零点间距60°。

M1	输入零点间距
A1	正确缩放和结论

Question 10

P2-C06-R10 · 6 marks

(a) · 3 marks

$x = -135^\circ, 45^\circ$.

解处cos不能为0，否则sin=±1不相等；可除cos得tanx=1，再筛选。

M1	除法合法说明或图像
M1	周期角
A1	两解

(b) · 3 marks

$-\sqrt{2}/2$ at -135°; $\sqrt{2}/2$ at 45°.

两比相等且平方和1，大小为$\sqrt{2}/2$，符号按象限。

M1	平方和或参考角
M1	象限符号
A1	两值

Question 11

P2-C06-R11 · 5 marks

(a) · 3 marks

$x = \pi/3, 2\pi/3.$

$\sin = \pm\sqrt{3}/2$，但该区间正弦非负，只选正值的两角。

M1	平方开根两符号
M1	结合区间筛选
A1	两角

(b) · 2 marks

$x = -2\pi/3, -\pi/3, \pi/3, 2\pi/3.$

扩展负半区间有负正弦对应的两个角，或平方函数关于0对称。

M1	新区间的负支
A1	全部角

Question 12

P2-C06-R12 · 6 marks

(a) · 3 marks

Multiply by $(1+\sin x)/(1+\sin x)$ and use $1-\sin^2 x=\cos^2 x$.

分子平方差变为$\cos^2$x，约分得到所给式；只在共同定义域讨论。

M1	共轭式
M1	平方恒等式
A1	正确简式

(b) · 3 marks

$\sqrt{3}/3$.

(1−1/2)/(√3/2)=1/√3，有理化。

M1	代精确三角值
M1	有理化
A1	值

Question 13

P2-C06-R13 · 5 marks

(a) · 3 marks

$60^\circ, 120^\circ, 240^\circ, 300^\circ$.

tan=$\pm\sqrt{3}$，共四象限的参考角60°。

M1	保留正负
M1	正切周期
A1	四解

(b) · 2 marks

$120^\circ, 240^\circ$.

余弦负在第二、第三象限，与(a)求交。

M1	正确象限筛选
A1	两角

Question 14

P2-C06-R14 · 6 marks

(a) · 3 marks

3, 2, 1, respectively.

零线包含区间两端和π共3点；中间正高度两点；最高点一。

B1	k零时
B1	中间值
B1	顶值

(b) · 3 marks

$-1 < k < 0$ or $0 < k < 1$.

正负内部高度均两点；k=0因端点变3，±1各1，范围外0。

M1	按高度范围分类
M1	处理边界与零
A1	完整区间

Question 15

P2-C06-R15 · 6 marks

(a) · 4 marks

$x = 45^\circ, 165^\circ$.

输入范围−30°到330°，cos为1/2的输入60°、300°；各加30再除2。

M1	正确输入范围
M1	筛选输入角
M1	还原x
A1	两解

(b) · 2 marks

45°.

最近的正输入解由−60°给x=−15°非法，60°给45°，故45为最小正。

M1	检查临近周期支
A1	最小值

Question 16

P2-C06-R16 · 7 marks

(a) · 3 marks

Use $\tan x = \sin x / \cos x$ and $\sin^2 x + \cos^2 x = 1$.

分子分母同乘$\cos^2 x$，分母变为$\sin^2 x+\cos^2 x=1$；原$\cos$不得为0。

M1	正切替换
M1	清分母及平方和
A1	目标式

(b) · 4 marks

$30^\circ, 150^\circ, 210^\circ, 330^\circ$.

$\sin^2=1/4$给$\sin=\pm 1/2$，四个角都不使$\cos=0$，所以均合法。

M1	平方开根
M1	全象限
M1	原域复核
A1	四解

Question 17

P2-C06-R17 · 5 marks

(a) · 2 marks

$1/4$.

用$\cos^2=1-\sin^2$，$1-\sin^2=\sin^2+1/2$。

M1	平方和替换
A1	结果

(b) · 3 marks

$\pi/6, 5\pi/6, 7\pi/6, 11\pi/6$.

$\sin=\pm 1/2$，在整周期列齐四个角。

M1	两符号
M1	弧度角筛选
A1	四解

Question 18

P2-C06-R18 · 6 marks

(a) · 2 marks

$x^2+y^2=25(\cos^2\theta+\sin^2\theta)=25.$

平方两坐标并用基本恒等式。

M1	正确平方相加
A1	常数25

(b) The point also lies on $y=\frac{5}{2}$. Find

(i) · 2 marks

$30^\circ, 150^\circ.$

5sinθ=5/2得sinθ=1/2，两解。

M1	正确三角方程
A1	两角

(ii) · 2 marks

$\pm 5\sqrt{3}/2.$

分别cos30=√3/2、cos150=−√3/2，乘5。

M1	象限符号
A1	两值

Question 19

P2-C06-R19 · 6 marks

(a) · 4 marks

$0^\circ, 90^\circ, 180^\circ, 360^\circ$.

sinx(sinx−1)=0，分别零点与最大值点；360端点包含。

M1	正确因式
M1	两种三角值
M1	端点
A1	解集

(b) · 2 marks

Missing $0^\circ, 180^\circ, 360^\circ$; division by $\sin x$ without checking its zeros.

除以可能为零的因子丢去整组零正弦解。

M1	指出除法问题
A1	全部漏解

Question 20

P2-C06-R20 · 6 marks

(a) · 3 marks

$x = -\arctan(1/3) \approx -0.322$.

tan在这一分支一一对应，负比值给负角，勿加π跑出域。

M1	正确反正切
M1	分支筛选
A1	精度

(b) · 3 marks

$3/\sqrt{10} = 3\sqrt{10}/10$.

参考直角三角形两直角边1、3，斜边$\sqrt{10}$；该区间cos正。

M1	参考边与勾股
M1	cos正号
A1	精确值

Question 21

P2-C06-R21 · 6 marks

(a) · 4 marks

$\sin x = (-2+\sqrt{29})/5$ only.

用cos²=1−sin²得5s²+4s−5=0，s=(−2±√29)/5；负号根小于−1须排。

M1	正确二次方程
M1	二次公式
M1	三角值范围筛选
A1	可行值

(b) · 2 marks

42.6° and 137.4°.

正弦值约0.677033，取arcsin及补角，按角度范围筛。

M1	反正弦与补角
A1	两角

Question 22

P2-C06-R22 · 6 marks

(a) · 4 marks

$x = \pi/3, \pi, 5\pi/3$.

令c=cosx，右边=1−2c²，得2c²+c−1=(2c−1)(c+1)=0。

M1	平方和化为余弦
M1	二次分解
M1	全部角
A1	三x

(b) · 2 marks

$1/2, -1, 1/2$ in that order.

在对应角上代cosx。

M1	使用同一图像函数值
A1	正确配对

Question 23

P2-C06-R23 · 6 marks

(a) · 4 marks

$x = 160/3, 340/3, 520/3, 700/3, 880/3, 1060/3$ degrees.

输入20°到1100°，正弦零点180°、360°、540°、720°、900°、1080°；减20再除3。

M1	正确输入区间
M1	完整输入零点
M1	还原x
A1	六解

(b) · 2 marks

3.

首三项约53.33、113.33、173.33；第四233.33已超出。

M1	正确筛选
A1	数量

Question 24

P2-C06-R24 · 7 marks

(a) find exactly

(i) · 3 marks

$-4/5$ and $-3/4$.

平方和给cos大小4/5，第二象限负；tan为二比之商。

M1	求cos大小
M1	象限负号
A1	两比

(ii) · 2 marks

4.

直接代(1+3/5)/(1−3/5)=(8/5)/(2/5)。

M1	正确分式代入
A1	值

(b) · 2 marks

$3/5$.

正弦在补角下相等，此为P1图像对称，不必用和角公式。

M1	补角对称
A1	精确值

Question 25

P2-C06-R25 · 6 marks

(a) · 3 marks

$1+3\cos^2 x$; minimum 1, maximum 4.

以1−cos²替换sin²，利用0⩽cos²⩽1。

M1	正确替换
M1	平方比范围
A1	极值

(b) · 3 marks

$60^\circ, 120^\circ, 240^\circ, 300^\circ$.

1+3cos²=7/4给cos²=1/4，四角。

M1	解平方三角值
M1	两符号与全象限
A1	四解

Question 26

P2-C06-R26 · 6 marks

(a) · 4 marks

$0, \pi/3, \pi, 5\pi/3, 2\pi.$

合法乘cosx得sinx(1−2cosx)=0；sin零给0、π、2π，cos1/2给π/3、5π/3，都满足原域。

M1	合法清分母
M1	因式分解
M1	全部角及原域
A1	解集

(b) · 2 marks

3.

删0和2π，保留三个内部解。

M1	端点删除
A1	数量

Question 27

P2-C06-R27 · 6 marks

(a) · 4 marks

$90^\circ, 120^\circ, 240^\circ, 270^\circ$.

cosx(2cosx+1)=0，分别cos0和−1/2。

M1	正确分解
M1	两种余弦值
M1	象限解
A1	完整集合

(b) · 2 marks

720°.

直接相加90+120+240+270。

M1	使用完整四解
A1	和

Question 28

P2-C06-R28 · 6 marks

(a) · 3 marks

$(\pi/12, 1/2)$ and $(5\pi/12, 1/2)$.

输入$2x\in[0,2\pi]$，等于$\pi/6$或$5\pi/6$，横坐标再除2。

M1	正确输入角
M1	横坐标缩放
A1	两点

(b) · 3 marks

$\pi/3$.

满足区间$[\pi/12,5\pi/12]$，长度$4\pi/12=\pi/3$。

M1	图像确定闭区间
M1	求区间长度
A1	结果

Question 29

P2-C06-R29 · 7 marks

(a) · 3 marks

$12/25.$

两边平方：1+2sinxcosx=49/25，故积12/25。

M1	展开平方与基本恒等式
M1	解乘积
A1	精确值

(b) · 4 marks

$\sin x = 3/5$ or $4/5$; $x \approx 36.9^\circ, 53.1^\circ$.

正弦与余弦的和7/5、积12/25，所以它们是$t^2-(7/5)t+12/25=0$的两根3/5、4/5；两锐角均合法。

M1	构造两比的二次方程
M1	解两比
M1	反三角与锐角范围
A1	两角

Question 30

P2-C06-R30 · 7 marks

(a) · 3 marks

$\cos\theta = (1 \pm \sqrt{41})/10.$

5(1−c²)+c=3给5c²−c−2=0；二次公式的判别式为41。

M1	正确平方替换
M1	二次求根
A1	两余弦值

(b) · 4 marks

$0.737, 2.142, 4.142, 5.546$ radians.

二次式根(1±√41)/10均在[−1,1]，每种余弦值对应两角。

M1	两个合法比值
M1	各两角
M1	弧度计算
A1	全部解